

XIAMEN UNIVERSITY

Programming Language C

Chapter 2

Types, Operators and Expressions (i) - Lab

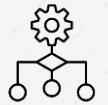
Week 2

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Outline

- 2.1 Basic C Data Types
- 2.2 Fundamental Operators
- 2.3 Operator Precedence
- 2.4 Expressions and Statements

2.1 Basic C Data Types



- 2.1.1 Types
- 2.1.2 Variables
- 2.1.3 Keywords
- 2.1.4 Identifiers
- 2.1.5 Constants/Literals

2.1.2 Variable

- Variables and Variable Definitions
  - Define Variables Before They Are Used
  - Defining Multiple Variables of the Same Type in One Statement
  - Identifiers and Case Sensitivity

- In C language, you declare a variable by specifying its data type followed by the variable name. Here's the basic syntax:

data_type variable_name

For example, if you want to declare an integer variable named **age**, you would write:

int age;

Practice--- Write C Program

- **Ex.1** State the order of evaluation of the operators in each of the following C statements and show the value of x after each statement is performed.

X = (8 + 15 * (6 – 2)) -1

X = 67

Practice--- Write C Program

- Ex.1 Solution

```c
#include <stdio.h>

int main(void) {
    // Declaring integer variables
    int a, b, c, x;

    // Initializing variables with arithmetic expressions
    a = 6 - 2;          // a = 4
    b = 15 * a;         // b = 15 * 4 = 60
    c = 8 + b;          // c = 8 + 60 = 68
    x = c - 1;          // x = 68 - 1 = 67

    // Displaying the values of variables using printf
    printf("The values are:\n");
    printf("a = %d\n", a);   // Prints the value of a
    printf("b = %d\n", b);   // Prints the value of b
    printf("c = %d\n", c);   // Prints the value of c
    printf("x = %d\n", x);   // Prints the value of x

    return 0;
}
```

Practice--- Write C Program

- Ex.2 State the order of evaluation of the operators in each of the following C statements and show the value of x after each statement is performed.

```
X = 5 % 5 + 5 * 5 – 5 / 5

X  =  24
```

Practice--- Write C Program

- Ex.2 Solution

```c
#include <stdio.h>

int main(void) {
    // Declaring integer variables
    int a, b, c, x;

    // Arithmetic operations with different operators
    a = 5 % 5;   // Modulus operation: 5 % 5 results in 0 (remainder when 5 is divided by 5)
    b = 5 * 5;   // Multiplication: 5 * 5 results in 25
    c = 5 / 5;   // Division: 5 / 5 results in 1 (integer division)

    // Perform a calculation using the values of a, b, and c
    x = a + b - c;   // Add 'a' and 'b', then subtract 'c'

    // Display the values of the variables
    printf("The results are:\n");
    printf("a = %d, b = %d, c = %d, x = %d\n", a, b, c, x);

    return 0;
}
```

Practice--- Write C Program

- Ex.3 State the order of evaluation of the operators in each of the following C statements and show the value of x after each statement is performed.

```
X = ( 5 * 7 * ( 5 + ( 7 * 5 / ( 7 ) ) ) )
```

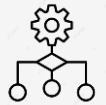
```
X  =  350
```

Practice--- Write C Program

- Ex.3 Solution

```
#include<stdio.h>
int main(void)
{
int x;
x=(5*7*(5+(7*5/(7))));
printf("%d",x);
}
```

Output Using “printf” in C



- **printf Function:**
  - Used to **display** output on the console.
  - Can also be used to **prompt the user for input** before using **scanf**.

- **Syntax:**
  - printf("Format String", arguments...);
  - Displays a message asking the user to provide input.

Format String: Contains **text** + **format specifiers** (%d, %f, etc.)

Arguments: Variables/values matching the format specifiers (**order matters**!)

Output Using “printf” in C

Common Format Specifiers:

Specifier	Usage	Example
%d	Integer	printf("%d", num);
%f	Float	printf("%.2f", pi);
%c	Character	printf("%c", ch);
%s	String	printf("%s", name);
%x	Hexadecimal	printf("%x", 255);

Output Using “printf” in C

Rules for Correct Usage

1.Number of specifiers == Number of arguments

- Wrong: printf("%d %f", x); (Missing argument for **%f**)
- Correct: printf("%d %f", x, y);

2.Type must match the specifier

- Wrong: printf("%d", 3.14); (**%d** expects int, but float given)
- Correct: printf("%f", 3.14);

3.Escape sequences work in format strings

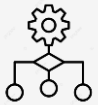
- \n (newline), \t (tab), \\ (backslash)
- **Example:** printf("Line 1\nLine 2");

Exercise 4

Task: Write a C program that does the following:

- **Declares** an integer variable **age** and assigns it the value **25**.
- **Declares** a floating-point variable **height** and assigns it the value **5.9**.
- **Uses printf** to display the output in the following format: **Age: 25, Height: 5.9 feet**
- The age should be printed as an integer (%d).
- The height should be printed as a float with 1 decimal place (%.1f).

Exercise 4 - Solution



```c
#include <stdio.h>
int main() {

    int age = 25;
    float height = 5.9;
    printf("Age: %d, Height: %.1f feet", age, height);
    return 0;

}
```

User Input - “scanf” Function

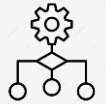
- **scanf Function:**
  - Used to **read formatted** input from the standard input (usually the keyboard).
  - **Stores** the entered data into specified variables.

- **Syntax:**
  - scanf("format_string", &variable1, &variable2, ...);
  - Displays a message asking the user to provide input.

Format String: Contains format specifiers **(%d, %f, etc.)** matching input type

Addresses (&): Memory locations where input will be stored (except for strings/arrays)

User Input - “scanf” Function



- **Critical Rules**
  - Always use **&** before variables (except for strings)
  - Match specifiers to variable types
  - Multiple inputs
- **Common Mistakes**
  - Missing &: Causes program crash
  - Type mismatch: Corrupts memory
  - Buffer overflow (for strings)

Exercise 5

Input/Output Operations

- **Task:** Write a complete C program that:
  - Declares an integer variable age and a floating-point variable height.
  - Prompts the user with the exact message: "**Enter age and height:** "
  - Reads **both values in a single scanf statement** (age first, then height).
  - Prints the output in this exact format: "**Age: [age], Height: [height]** "
  - Display height with 1 decimal place.
- **Requirements:**
  - Use %d for age and %f for height in both scanf and printf.

Exercise 5 - Solution

```c
#include <stdio.h>
int main() {
    int age;
    float height;
    printf("Enter age and height: ");
    scanf("%d %f", &age, &height);  // Reads integer + float
    printf("Age: %d, Height: %.1f", age, height);
    return 0;
}
```

Practice--- Write C Program

- Ex.6 Write a program that asks the user to enter the initial velocity and acceleration of an object, and the time that has elapsed, places them in the variables u, a, and t, and prints the final velocity, v, and distance traversed, s, using the following equations.

$$v = u + at$$

Practice--- Write C Program

- Ex.6 Solution

```c
#include <stdio.h>

int main(void) {
    // Declare integer variables
    int u, a, t, v;

    // Input values for u, a, and t from the user
    scanf("%d %d %d", &u, &a, &t);

    // Compute the value of v based on the formula: v = u + a * t
    v = u + a * t;

    // Output the value of v
    printf("v is %d", v);

    return 0;
}
```

Practice--- Write C Program

- Ex.7 Write a program than asks the user to enter the initial velocity and acceleration of an object, and the time that has elapsed, places them in the variables u, a, and t, and prints the final velocity, v, and distance traversed, s, using the following equations.

$$s = ut + \frac{1}{2}at^2$$

Practice--- Write C Program

- Ex.7 Solution

```c
#include <stdio.h>

int main(void) {
    // Declare variables
    float a, t, v, s;

    // Prompt user to input values for acceleration, time, and initial velocity
    printf("Enter acceleration (a), time (t), and initial velocity (v):\n");

    // Input values
    scanf("%f %f %f", &a, &t, &v);

    // Calculate displacement
    s = v * t + 1.0 / 2.0 * a * t * t;

    // Print the result
    printf("Displacement (s) is %f\n", s);

    return 0;
}
```

Exercises

1. Write a C program that calculates the volume of a cylinder using the formula:

$$V = \pi . r^2 . h$$

   - Where:
     - V is the volume,
     - r is the radius of the base, and
     - h is the height of the cylinder.

   - Use 3.14159 as the value for π (Constant). The program should take the radius (r) and height (h) as inputs and calculate the volume

2. Write a C program that takes three integers as input from the user and performs the following operations:
   - Calculate the modulus of the first number by the second.
   - Multiply the second number by the third.
   - Divide the first number by the third.

3. Write a C program that converts a temperature from Celsius to Fahrenheit using the formula: $F = \frac{9}{5} \cdot C + 32$

   Where:

   - F is the temperature in Fahrenheit,
   - C is the temperature in Celsius.

   The program should prompt the user to input the temperature in Celsius and output the corresponding temperature in Fahrenheit.

Exercises

4. Write a C program to solve for a in the following equation: $b = \frac{a + c}{d}$

   - Rearrange the equation to solve for a: $a = b \cdot d - c$

     Your program should prompt the user to input values for b, c, and d, and then calculate and display the value of a.



Thank you

Q&A